



Deformation behavior of fluid-filled porous elastomers: Analytical estimates and validation

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ABSTRACT

Fluid-filled elastomers are used in soft body wearable electronics devices, optics, soft pressure transducers, vibration damping and impact protection. Deformation behavior of these materials is complex and would depend on elastomer properties, fluid properties, loading rate, porosity, pore size, pore shape, and pore end conditions. In this work, closed form analytical estimates are derived for effective behavior of a special class of fluid-filled elastomers composed of incompressible, hyperelastic matrix with mono-disperse circular pores, distributed in a square grid, filled with an incompressible, Newtonian fluid. The analytical estimates are validated using three dimensional numerical simulations modeled in COMSOL multiphysics software. The analytical estimates for pressure and velocity fields in the fluid, deformation of the matrix, viscous dissipation in fluid, and stored energy in the solid are seen to match well with that from numerical simulations. For small deformations and small porosities, while storage modulus is seen to be linearly proportional to shear modulus of elastomer and inversely proportional to the porosity, the loss modulus is seen to be linearly proportional to the product of loading frequency and fluid viscosity and inversely proportional to the product of porosity and square of ratio of pore radius to pore length. The analytical estimates presented here can be used to design fluid-filled elastomers with desired stiffness and damping by selecting appropriate elastomer stiffness, fluid viscosity, porosity, pore size and pore aspect ratio.

1. Introduction

Vibration damping and impact mitigation are needed in automobiles (Zhang and Dupuis, 2011; Deng et al., 2003), bullet proof vests (Bazhenov, 1997), protection equipment (Ramirez and Gupta, 2018), sports equipment (Carré et al., 2004), body wearable devices (Ma et al., 2017), pressure transducers (Park et al., 2010) and optical lenses (Shian et al., 2013). Fluid-filled porous elastomers, with continuous inter-connected pore networks can be used in such applications. The fluid in the pores squeezes out of the elastomeric matrix when such materials (e.g. cartilage, body tissue, sponges, etc.) are compressed and the fluid gets sucked in when the compressive load is removed. The fluid dissipates energy as it flows in and out of the pores. In view of the enormous potential they hold, it is essential to know their effective behavior as a function of the elastomer properties, fluid properties, pore microstructure, pore end conditions, and loading rate. Coupled fluid–structure interaction (FSI) occurs in fluid-filled elastomers as they are subjected to deformation. Closed form analytical estimates for macroscopic behavior of fluid-filled elastomers as a function of pore microstructure, loading rate, elastomer, and fluid properties have not been presented till date. This work aims to address this lacuna.

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Porous elastomers (without any fluid in them) have wide range of applications including that in packaging, automobiles, furniture, energy absorption and noise reduction. Macroscopic behavior of porous elastomers is complex and depends on nature of loading, elastomer properties, and pore microstructure (porosity, pore size, pore shape, pore size distribution, open/closed cell nature of pores, etc.). Numerous efforts have been made to develop constitutive models that are simple, yet capable of modeling the macroscopic behavior of porous elastomers. [Blatz and Ko \(1962\)](#) carried out experiments on polyurethane rubbers having 50% volume fraction of randomly distributed pores of 40 μm diameter and proposed a phenomenological model for the same. Though their material model could only predict the deformation behavior in a limited manner, it could predict the onset of macroscopic instability at large deformations ([Knowles and Sternberg, 1975](#)). A number of homogenization based models have been proposed ([Ogden, 1978](#); [Feng and Christensen, 1982](#); [Hashin, 1985](#); [Danielsson et al., 2004](#)) for describing the constitutive behavior of porous elastomers. In particular, [Danielsson et al. \(2004\)](#) proposed an estimate for overall response of porous elastomers having incompressible, isotropic matrix phase. The proposed estimate was an upper bound and was obtained assuming a composite spherical assemblage microstructure. [Guo et al. \(2008\)](#) proposed a hyperelastic constitutive model for a neo-Hookean elastomer having aligned continuous cylindrical pores based on multiplicative decomposition of applied macroscopic deformation gradient. On lines of models for porous elastomers, a number of constitutive models have been proposed for porous viscoelastic materials ([Nguyen et al., 2016](#); [Xu et al., 2007](#); [Singh et al., 2021](#)). In elastomers, the external work is stored in the form of strain energy during loading and stored energy is released upon unloading. For viscoelastic materials, the external work is partially stored and partially dissipated. While there are several works on constitutive models characterizing the macroscopic deformation behavior of porous elastomers and porous viscoelastic composites, there are no earlier studies, characterizing the macroscopic response of fluid-filled elastomers through a micro-mechanics or homogenization based approach, that relate elastomer properties, pore microstructure, fluid properties, and loading rate to the resulting macroscopic stress, viscous dissipation, and strain energy. Nonetheless, there are some earlier works on experimental characterization of fluid-filled elastomers ([Hilyard, 1971](#); [Hilyard and Kanakkanatt, 1970](#); [Rehkopf et al., 1996](#); [Dawson et al., 2008, 2009](#); [Yost, 2012](#); [Stewart et al., 2010](#); [Vechart, 2011](#)) and mathematical modeling of polymeric gels ([Hong et al., 2008](#)) using poroelasticity theory ([Biot, 1941](#)).

Use of open-cell foams fully saturated with Newtonian or non-Newtonian fluids for energy absorption and impact mitigation has been studied (primarily through experiments) to a limited extent earlier. [Hilyard \(1971\)](#) carried out one of the earliest studies on open cell elastomeric foams fully saturated with incompressible fluids. Liquids considered were water and 70% glycerine–water solution. The foam was kept submerged in fluid. Fluid flow velocity was considered to be perpendicular to the direction of impact (i.e. since impact was in vertical direction flow velocity was in horizontal direction). Fluid flow velocity was maximum at the top surface and minimum at bottom surface. Transverse deformation of the matrix was also considered. Experiments were carried out for a fluid-filled foam of dimensions 4 cm $\times$ 4 cm $\times$ 3 cm, with one of the sides open to a fluid reservoir at atmospheric pressure. All sides except one lateral surface were coated with latex. Equations of motion for the mass striking the fluid-filled foam were solved numerically. The results for energy absorption were compared with that from experiments. Impact tests were carried out for two types of foams saturated with fluid. It was observed that the damping capacity initially increased with the velocity of impacting body, reached a maximum and then decreased. Maximum damping capacity increased with the mass of the impacting body. Since fluid is one of the phases in the composite, operating temperature and loading frequency have a significant effect on their response. Therefore going ahead ([Hilyard and Kanakkanatt, 1970](#)) carried out experiments to study fluid flow induced damping in open-cell polyurethane foam saturated with silicone oil and glycerine over a wide range of temperatures and frequencies. Experimental results were compared with Gent–Rusch model ([Gent and Rusch, 1966](#)). It was observed that for temperatures well above glass transition temperature of the polymer, damping for fluid-filled foam was independent of temperature. [Rehkopf et al. \(1996\)](#) carried out experiments to identify the contribution of solid and fluid to overall behavior of closed cell foams under dynamic loading conditions. The foams used were polyurethane furniture foam and semi-rigid polyurethane foam. The foams had ribbed microstructure. The fluid in this case was air. Experiments were done to study the effect of trapped air on the response of the foams.

Fatal injuries from accidents and war were a motivation for creating helmets and protection equipment that mitigate impacts and protect body and brain. A number of experiments were carried out to evaluate the performance of porous elastomers filled with fluids under dynamic loading. [Dawson et al. \(2008, 2009\)](#) investigated the response of a deformable open-cell polyurethane foam fully saturated with a highly viscous Newtonian fluid and a colloidal silica based non-Newtonian fluid under dynamic loading for use in helmets. One dimensional fluid flow problem was considered assuming Stokes flow. Lubrication approximation ([Batchelor, 2000](#)) that considers pore cross-sectional dimensions to be much smaller than the pore axial length was taken. Multiple combinations of foam and filler materials were tested under dynamic loading to evaluate their performance in preventing brain injuries. Experimental investigations of fluid-filled closed cell vinyl nitrile (VN) foams for use as helmet liners, for protection against traumatic brain injury, were carried out by [Yost \(2012\)](#). Channels were made in the foam, which were filled with glass beads or glycerine. The peak pressure transmitted by filled foams was compared with that of solid foam. [Stewart et al. \(2010\)](#) conducted drop tests on conventional expanded polystyrene and unsaturated VN foams to compare their energy absorption capacities, under constrained and unconstrained end conditions. Fluid-filled VN foams, with unconstrained fluid channels, were found to best reduce the effect of impact. [Vechart \(2011\)](#) observed that glycerine filled elastomeric liner performed better than the conventional combat helmet liner.

In this paper, for incompressible elastomers containing mono-dispersed circular pores fully saturated with a fluid, analytical estimates are derived for the following: (i) average deformation gradient and average strain energy density in the solid phase, (ii) pore size and shape evolution as a function of macroscopic deformation, (iii) velocity and pressure fields in the fluid within the pores of the elastomer, (iv) viscous dissipation in the fluid from deformation induced flow, (v) overall response of the fluid-filled elastomer and (vi) equivalent spring stiffness and damping coefficient for the fluid-filled elastomers subjected to plane strain compression. Analytical estimates are compared with numerical simulations modeled in COMSOL multiphysics software to verify their accuracy.

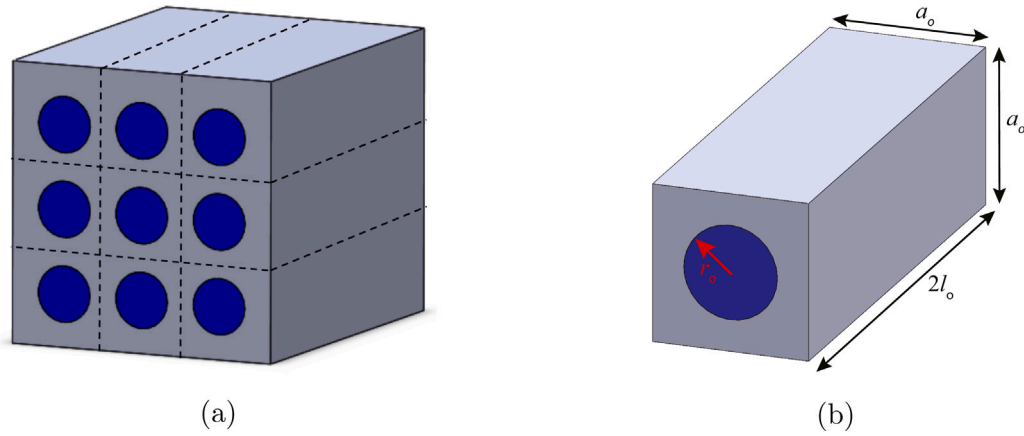


Fig. 1. Schematic of porous elastomers with square arrangement of cylindrical pores. (a) Mono-dispersed pores arranged in a square array in an elastomer. The pores are fully saturated with a fluid. (b) Unit cell or representative volume element (RVE) of this microstructure.

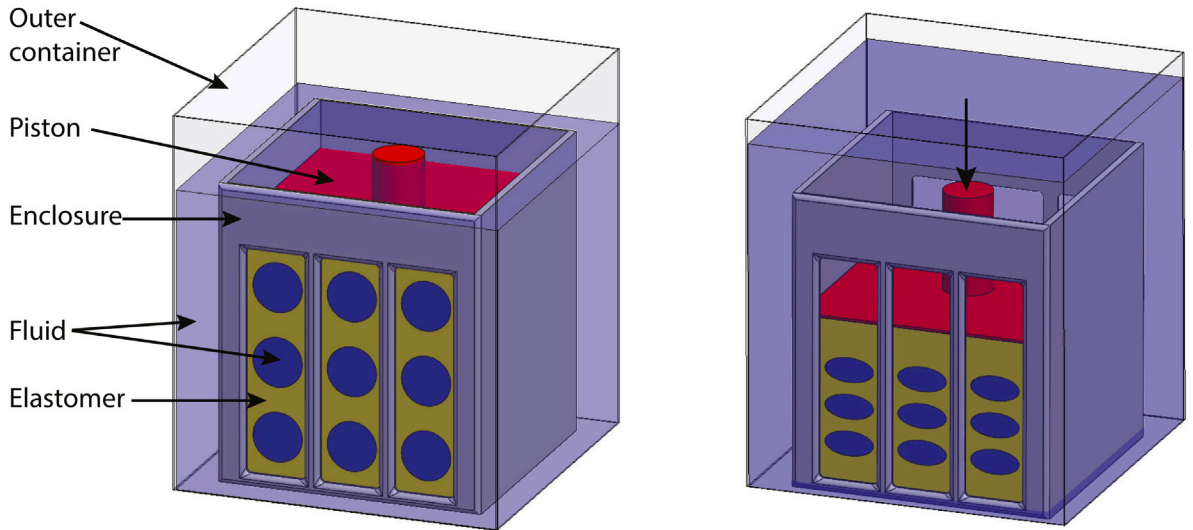


Fig. 2. Schematic of the fluid-filled elastomer subjected to uniaxial plane strain loading. The figure on the left shows elastomer in undeformed configuration, while that on the right shows it in deformed configuration. The fluid level in outer container rises as the piston moves down during compression.

Two main concepts are employed to obtain analytical estimates for average deformation gradient, stored energy in solid phase and pressure, velocity fields in the fluid phase. The stored energy and average deformation gradient in solid phase are determined by solving an equivalent problem for cylindrical assemblage with identical porosity. The velocity and pressure fields for the fluid are obtained using a weighted integral approach. A flowchart showing the methodology adopted for estimation of strain-energy and average deformation gradient in the elastomer, average deformation gradient in the pores, pressure, velocity, dissipation density for the fluid, and overall response for the fluid filled elastomer is shown in Fig. 4.

2. Problem formulation

2.1. Geometry construction

In the present work, the foam microstructures shown in Fig. 1(a) is considered. The unit cell geometry considered in analytical studies and COMSOL simulations, for microstructure with square arrangement of pores, is shown in Fig. 1(b). Pore initial radius r_o , initial length $2l_o$ and initial porosity f_o characterize the geometry of the unit cell.

Fig. 2 shows a schematic of the fluid-filled elastomer subjected to uniaxial plane strain loading for which closed form analytical results are derived. It may be noted that the fluid-filled elastomer works as simple spring-damper system in this case. The elastomer is kept in an enclosure with vertical slits such that when subjected to compression using a piston, the elastomer experiences uniaxial

plane strain compression. The entire assembly is kept in an outer container filled with the same fluid as in the pores. As the piston compresses the composite, the pores reduce in size and the fluid in the pores comes out into the outer container. During unloading, when the piston moves up, the pores enlarge and the fluid moves from the outer container into the pores.

2.2. Material behavior

2.2.1. Fluid modeling

A Newtonian, incompressible fluid is taken to fully occupy the pores. Incompressible Navier–Stokes equation is enforced in the fluid domain. More specifically, the continuity equation

$$\nabla \cdot \mathbf{v} = 0 \quad (1)$$

and the Navier–Stokes equation

$$\rho_f \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \nabla \cdot [\mu (\nabla \mathbf{v})_{\text{sym}}] \quad (2)$$

are enforced in the fluid domain. Here, p is pressure, $\mathbf{v}$ is velocity vector, ρ_f is fluid density and μ is fluid viscosity. Velocity and pressure are the primary variables that need to be determined for the fluid phase. Only the fluids that do not chemically interact/degrade/swell the elastomers are considered in this work. For example, glycerol could be the fluid and silicone the elastomer for the system considered here.

2.2.2. Elastomer modeling

The solid phase in the composite is modeled as an incompressible, Neo-Hookean hyperelastic material. Its strain energy density is taken as

$$W^{(s)}(\mathbf{F}) = \frac{G}{2} (I_1 - 3), \quad (3)$$

where G denotes the shear modulus, $I_1 = \text{tr}(\mathbf{C})$ is the first principal invariant of right Cauchy–Green deformation tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and $\mathbf{F}$ is the deformation gradient tensor. Neglecting inertial effects and body forces, the equilibrium equation for the solid phase is taken as

$$\nabla \cdot \boldsymbol{\sigma} = 0, \quad (4)$$

where

$$\boldsymbol{\sigma} = \frac{1}{\det(\mathbf{F})} \mathbf{F} \frac{\partial W^{(s)}}{\partial \mathbf{F}} \quad (5)$$

is the Cauchy stress tensor. Displacements are primary variables for the solid phase.

2.3. Boundary conditions

2.3.1. Fluid domain

While no slip boundary condition is assumed for the fluid on pore inner walls, constant pressure condition is assumed at pore ends where the fluid drains into the outer container (see Fig. 2). Fluid flow is induced by the deformation of the elastomer. It may be noted that while constant pressure at pore ends is relevant only to the fluid, no slip boundary condition on pore inner surface is applicable to the solid and the fluid phases.

2.3.2. Solid domain

Affine boundary conditions are applied on the unit cell/representative volume element (RVE) boundary (only on outer boundary and not the inner), i.e. displacements on the RVE outer boundary, for the elastomer phase, are taken as

$$\mathbf{u} = (\bar{\mathbf{F}} - \mathbf{I})\mathbf{X}, \quad (6)$$

where $\bar{\mathbf{F}}$ is the macroscopic deformation gradient tensor. Here, analytical estimates are obtained for in-plane deformations of the form

$$[\bar{\mathbf{F}}] = \begin{bmatrix} \bar{\lambda}_1 & 0 & 0 \\ 0 & \bar{\lambda}_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (7)$$

Presence of pores in the elastomer makes the composite compressible (from change of volume of pores) although the matrix itself is incompressible, therefore, $\det \bar{\mathbf{F}} \neq 1$. Fluid-filled elastomers, having in-compressible solid and fluid phases, exhibit compressible behavior for finite loading rates if the fluid is allowed to go out or come into the material system. Flow of fluid, in and out of pores, gives the fluid-filled elastomer the ability to dissipate energy as it is deformed. Thus, in this work, finite loading rates and constant pressure end condition for pores, are considered.

Table 1
Mesh parameters for different mesh sizes.

Parameters	Coarse mesh	Normal mesh	Fine mesh
No. of domain elements	98 791	190 623	503 368
No. of boundary elements	16 050	24 530	46 540
No. of edge elements	1140	1432	1966
Min. solid element size (mm)	0.35	0.225	0.125
Max. solid element size (mm)	1.88	1.25	1.0
Min. fluid element size (mm)	0.0252	0.0189	0.0126
Max. fluid element size (mm)	0.0818	0.0629	0.0422
Boundary layers	2	2	2

2.4. Numerical modeling

Coupled fluid–structure interaction simulations are carried out in COMSOL multiphysics software. The solid domain is defined using the solid mechanics module and the fluid domain is defined using the laminar flow module with inertial terms. The two modules are connected by fluid–structure interaction multiphysics module. Since the fluid domain has a moving boundary its mesh deforms during the simulation. Grid independence is checked by solving the problem for different mesh sizes. Grid independence tests are carried out for initial porosity $f_o = 0.6$ and initial pore radius $r_o = 0.5$ mm. Properties of the three meshes considered are shown in Table 1.

Solid domain is discretized using quadratic serendipity elements. The serendipity element has nodes on the vertices and edge centers. The serendipity elements provide good accuracy at relatively lower computational cost. Fluid domain is discretized using P2+P1 scheme, i.e. second-order elements for the velocity components and linear elements for the pressure field. Backward differentiation formula (BDF), an implicit method has been used for time stepping. PARDISO, a direct solver available within COMSOL multiphysics was used. For convergence, the relative tolerance was set at 0.005. In numerical analysis, Eulerian formulation is used for the fluid, while Lagrangian formulation is used for the solid. An arbitrary Lagrangian–Eulerian method is used to handle the fluid–structure interaction. Velocity continuity is enforced at pore inner surface.

3. Analytical estimates

3.1. Deformation of elastomeric phase

Average deformation gradient in the solid phase is taken to depend only on the initial porosity and applied macroscopic deformation gradient. More specifically it is taken to be identical to that in a cylindrical assemblage with identical porosity. The average deformation gradient in the solid phase in cylindrical assemblage is estimated by solving for deformation in annular region shown in Fig. 3(b) for affine boundary conditions prescribed in Eq. (6), i.e. the circular annular regions are subjected to macroscopic deformation represented by principal (generally unequal) macroscopic stretches $(\bar{\lambda}_1, \bar{\lambda}_2, \bar{\lambda}_3 = 1)$. The applied macroscopic deformation gradient $\bar{\mathbf{F}}$, is taken to be of the form

$$[\bar{\mathbf{F}}] = \begin{bmatrix} \bar{\lambda}_1 & 0 & 0 \\ 0 & \bar{\lambda}_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (8)$$

where δ_{ij} is kronecker delta. The initial pore volume fraction f_o is related to geometric parameters of annular regions in cylindrical assemblage through relation

$$f_o = (R_i/R_o)^2, \quad (9)$$

where R_i and R_o are initial inner and outer radii of a circular annular region, respectively. The outer boundary of the annular region changes to an elliptical shape, in the deformed configuration, as the cylindrical assemblage is subjected to $\bar{\mathbf{F}}$, as shown in Fig. 3(c).

A kinematically admissible deformation field, on lines of that proposed in Hang-Sheng and Abeyaratne (1992) for spherical assemblage, is assumed such that

$$x_i = \psi_i(R) X_i, \quad R = \sqrt{X_1^2 + X_2^2} \quad (i = 1, 2) \quad \text{and there is no sum on } i \quad (10)$$

where $\mathbf{x}$ denotes the position of a material point in the solid phase in deformed configuration, $\mathbf{X}$ denotes the position of the corresponding material point in initial/undeformed configuration, and R is the radial distance of the point from pore center in the undeformed configuration. Here, $x_3 = X_3$. The components X_i ($i = 1, 2$) in cylindrical coordinate system are given as

$$X_1 = R \cos \theta, \quad X_2 = R \sin \theta \quad (11)$$

where $0 \leq \theta \leq 2\pi$. Components of the local deformation gradient in the elastomeric matrix can be obtained as

$$\mathbf{F}_{ij}^{(s)} = \psi_i' \frac{X_i X_j}{R} + \psi_i \delta_{ij}, \quad i, j = 1, 2 \quad (\text{no sum on } i) \quad (12)$$

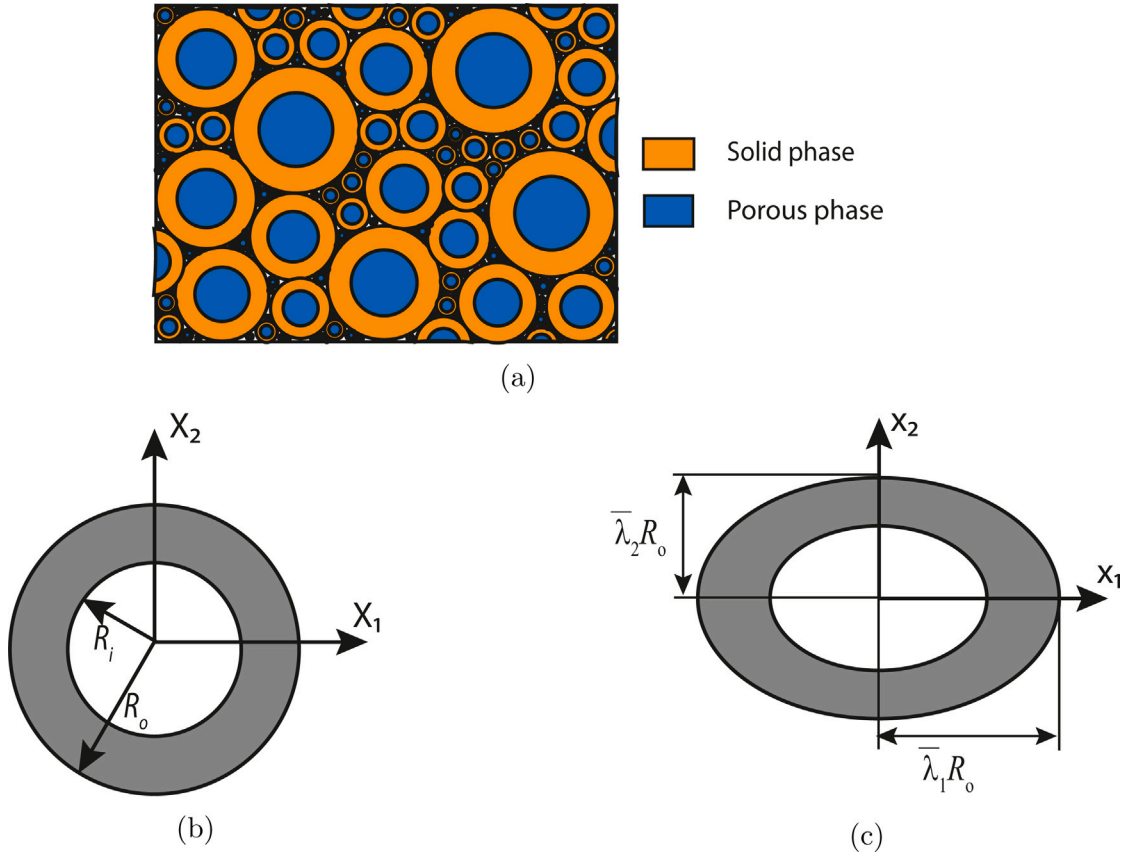


Fig. 3. (a) Cylindrical assemblage considered for estimation of average deformation gradient in solid phase, (b) undeformed and (c) deformed configurations of an annular region in the cylindrical assemblage. All annular regions in the cylindrical assemblage are subjected to the same macroscopic deformation gradient $\bar{\mathbf{F}}$.

where the $()'$ symbol denotes derivative with respect to R . The elastomer is taken to be locally incompressible, therefore,

$$J^{(s)} \equiv \det \mathbf{F}^{(s)} = 1 \quad (13)$$

From Eq. (13), it can be shown that at all points in the solid phase

$$\begin{aligned} \psi_1 \psi_2 + R \psi_1 \psi_2' &= 1 \quad \text{and} \\ \psi_1' \psi_2 - \psi_1 \psi_2' &= 0 \end{aligned} \quad (14)$$

The two unknown functions ψ_1 and ψ_2 can be obtained by solving the system of differential equations in Eq. (14). To solve for these, we use the boundary conditions. The circular outer surface (with radius R_o) in initial configuration transforms into an ellipse with semi axes $\bar{\lambda}_1 R_o$, $\bar{\lambda}_2 R_o$ under the applied macroscopic deformation $\bar{\mathbf{F}}$. Using Eq. (10), we get

$$\psi_i|_{R=R_o} = \bar{\lambda}_i \quad (i = 1, 2) \quad (15)$$

Using Eq. (15) and system of equations in Eq. (14), it can be shown that

$$x_i = \frac{\bar{\lambda}_i}{\bar{J}^{1/2}} \left[1 + \left(\frac{R_o}{R} \right)^2 (\bar{J} - 1) \right]^{1/2} X_i, \quad (i = 1, 2) \quad (\text{no sum on } i) \quad (16)$$

where $\bar{J} = \bar{\lambda}_1 \bar{\lambda}_2$ and $x_3 = X_3$. Using Eqs. (12) and (16), the components of the local deformation gradient tensor $\mathbf{F}^{(s)}$ in the elastomeric matrix, can be expressed as

$$\begin{aligned} F_{ij}^{(s)} &= \frac{\bar{\lambda}_i}{\bar{J}^{1/2}} \left(\frac{1 - \psi^2}{\psi R^2} X_i X_j + \psi \delta_{ij} \right) \quad (i, j = 1, 2) \quad (\text{no sum on } i) \\ \bar{F}_{31}^{(s)} &= \bar{F}_{13}^{(s)} = \bar{F}_{23}^{(s)} = \bar{F}_{32}^{(s)} = 0, \quad \bar{F}_{33}^{(s)} = 1 \end{aligned} \quad (17)$$

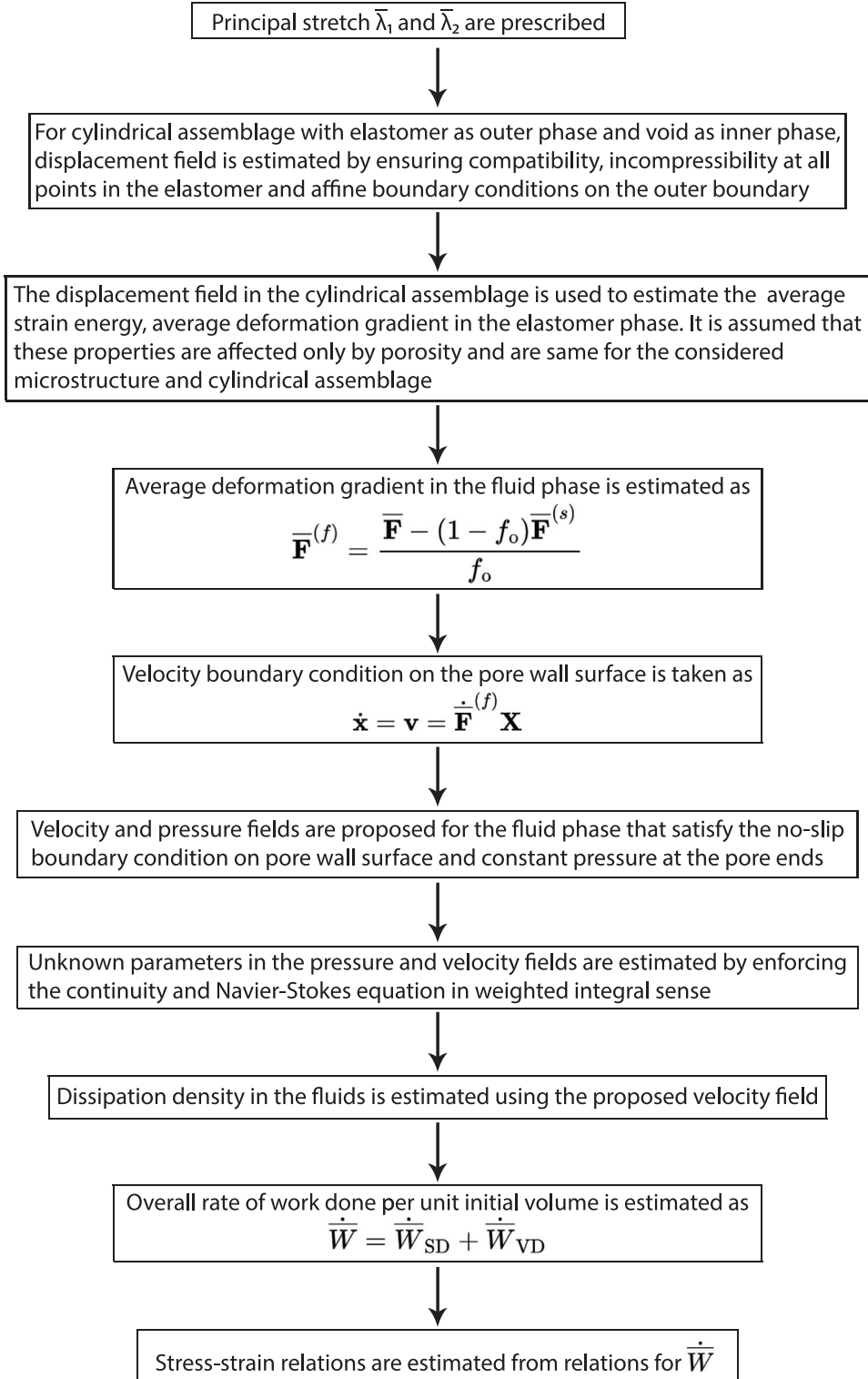


Fig. 4. Flowchart showing steps adopted in deriving the analytical estimates for macroscopic response of fluid-filled elastomer.

where $\psi = \left[1 + (R_o/R)^2 (\bar{J} - 1)\right]^{1/2}$. Using Eqs. (12) and (16), the first invariant of right Cauchy–Green deformation tensor can be obtained as

$$I_1^{(s)} = \frac{1}{J} \left\{ (\bar{I}_1 - 1) \psi^2 + \frac{1}{R^2} \left(\frac{1}{\psi^2} - \psi^2 \right) (\bar{\lambda}_1^2 X_1^2 + \bar{\lambda}_2^2 X_2^2) \right\} + 1 \quad (18)$$

The average strain energy density for the porous solid $\bar{W}_{SD}$ is obtained by integrating the pointwise strain energy density function W over the volume of circular annular region and then dividing it by the reference volume, i.e.

$$\bar{W}_{SD} = \frac{1}{\pi R_o^2 2l_o} \int_{-l_o}^{l_o} dz \int_{R_i}^{R_o} R dR \int_0^{2\pi} W^{(s)} d\theta \quad (19)$$

Using Eqs. (19) and (3) the macroscopic strain energy density of the porous elastomer is obtained as

$$\bar{W}_{SD} = \frac{G}{2} \frac{(\bar{I}_1 - 1)(\bar{J} - 1)}{2\bar{J}} \log \left(\frac{f_o + \bar{J} - 1}{f_o \bar{J}} \right) + G(1 - f_o) \left(\frac{(\bar{I}_1 - 1)}{2\bar{J}} - 1 \right) \quad (20)$$

The average deformation gradient in the solid phase is estimated as

$$\bar{\mathbf{F}}^{(s)} = \frac{1}{\pi(R_o^2 - R_i^2)} \int_{R_i}^{R_o} R dR \int_0^{2\pi} \mathbf{F}^{(s)} d\theta \quad (21)$$

The evolution of pore shape and size is governed by the average deformation gradient of the porous (fluid) phase $\bar{\mathbf{F}}^{(f)}$. The microstructural variables characterizing the pore size and shape are the current pore volume fraction f and the current pore aspect ratio ω . $\bar{\mathbf{F}}^{(f)}$ is computed by making use of the condition

$$\bar{\mathbf{F}} = (1 - f_o) \bar{\mathbf{F}}^{(s)} + f_o \bar{\mathbf{F}}^{(f)} \quad (22)$$

Using Eqs. (21) and (22) the components of $\bar{\mathbf{F}}^{(f)}$ can be estimated as

$$\begin{aligned} \bar{F}_{11}^{(f)} &= \bar{\lambda}_1^{(f)} = \bar{\lambda}_1 \sqrt{\frac{(f_o + \bar{J} - 1)}{f_o \bar{J}}} \\ \bar{F}_{22}^{(f)} &= \bar{\lambda}_2^{(f)} = \bar{\lambda}_2 \sqrt{\frac{(f_o + \bar{J} - 1)}{f_o \bar{J}}} \\ \bar{F}_{12}^{(f)} &= \bar{F}_{21}^{(f)} = \bar{F}_{31}^{(f)} = \bar{F}_{13}^{(f)} = \bar{F}_{23}^{(f)} = \bar{F}_{32}^{(f)} = 0, \quad \bar{F}_{33}^{(f)} = \bar{\lambda}_3^{(f)} = 1 \end{aligned} \quad (23)$$

It follows from Eq. (23) that

$$\det \bar{\mathbf{F}}^{(f)} = \frac{\bar{\lambda}_1^{(f)} \bar{\lambda}_2^{(f)} \bar{\lambda}_3^{(f)}}{\bar{\lambda}_1 \bar{\lambda}_2 \bar{\lambda}_3} = \frac{\bar{J} + f_o - 1}{f_o}. \quad (24)$$

The current volume fraction of pores f and current aspect ratio of pores ω can be obtained as

$$f = \frac{\det \bar{\mathbf{F}}^{(f)}}{\det \bar{\mathbf{F}}} f_o = \frac{\bar{\lambda}_1^{(f)} \bar{\lambda}_2^{(f)}}{\bar{\lambda}_1 \bar{\lambda}_2} f_o = \frac{f_o + \bar{J} - 1}{\bar{J}}, \quad \bar{J} = \bar{\lambda}_1 \bar{\lambda}_2 \quad (25)$$

and

$$\omega = \frac{\bar{\lambda}_2^{(f)}}{\bar{\lambda}_1^{(f)}} = \frac{\bar{\lambda}_2}{\bar{\lambda}_1}. \quad (26)$$

3.2. Pressure and velocity fields in the fluid domain

The pore boundary is taken to be subjected to affine displacement $\mathbf{u}$ of the form

$$\mathbf{u} = \bar{\mathbf{F}}^{(f)} \mathbf{X} - \mathbf{X} \quad (27)$$

Assuming no-slip condition on pore boundaries, fluid velocity $\mathbf{v}$ on the pore boundaries is estimated as

$$\mathbf{v} = \frac{d\mathbf{u}}{dt} = \dot{\bar{\mathbf{F}}}^{(f)} \bar{\mathbf{F}}^{(f)-1} \mathbf{x} = \bar{\mathbf{L}}^{(f)} \mathbf{x}, \quad (28)$$

where

$$\bar{\mathbf{L}}^{(f)} = \begin{bmatrix} \dot{\bar{\lambda}}_1^{(f)} / \bar{\lambda}_1^{(f)} & 0 & 0 \\ 0 & \dot{\bar{\lambda}}_2^{(f)} / \bar{\lambda}_2^{(f)} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad (29)$$

with

$$\bar{\lambda}_1^{(f)} = \bar{\lambda}_1 \sqrt{\frac{(f_0 + \bar{J} - 1)}{f_0 \bar{J}}}, \quad \dot{\bar{\lambda}}_1^{(f)} = \frac{2\dot{\bar{\lambda}}_1 \bar{J}(f_0 + \bar{J} - 1) + \bar{\lambda}_1(1 - f_0)\dot{\bar{J}}}{2f_0^{1/2} \bar{J}^{3/2} \sqrt{f_0 + \bar{J} - 1}}, \quad (30)$$

$$\bar{\lambda}_2^{(f)} = \bar{\lambda}_2 \sqrt{\frac{(f_0 + \bar{J} - 1)}{f_0 \bar{J}}}, \quad \dot{\bar{\lambda}}_2^{(f)} = \frac{2\dot{\bar{\lambda}}_2 \bar{J}(f_0 + \bar{J} - 1) + \bar{\lambda}_2(1 - f_0)\dot{\bar{J}}}{2f_0^{1/2} \bar{J}^{3/2} \sqrt{f_0 + \bar{J} - 1}}. \quad (31)$$

Here $\bar{\lambda}_1^{(f)}, \bar{\lambda}_2^{(f)}, \bar{\lambda}_3^{(f)} = 0$ are the average stretch rates for the pore along X_1, X_2 and X_3 directions. Using Eqs. (28) and (29) the velocity components for material points on pore boundary can be estimated as

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} \dot{\bar{\lambda}}_1^{(f)} x_1 / \bar{\lambda}_1^{(f)} \\ \dot{\bar{\lambda}}_2^{(f)} x_2 / \bar{\lambda}_2^{(f)} \\ 0 \end{pmatrix} \quad (32)$$

Considering Eq. (32), kinematically admissible velocity and pressure fields in the fluid domain are proposed as

$$v_1 = \frac{\dot{\bar{\lambda}}_1^{(f)} x_1}{\bar{\lambda}_1^{(f)}} \left(A \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_o^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_o^2} \right) + 1 \right) \quad (33a)$$

$$v_2 = \frac{\dot{\bar{\lambda}}_2^{(f)} x_2}{\bar{\lambda}_2^{(f)}} \left(B \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_o^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_o^2} \right) + 1 \right) \quad (33b)$$

$$v_3 = C \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_o^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_o^2} \right) \frac{x_3}{l_o} \quad (33c)$$

$$p = \left(1 + D \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_o^2} + E \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_o^2} \right) \left(1 - \frac{x_3^2}{l_o^2} \right) F + p_o \quad (33d)$$

The proposed velocity and pressure fields satisfy the following conditions:

1. Pressure is equal to p_o at pore openings ($x_3 = \pm l_o$)
2. Axial velocity $v_3 = 0$ on $x_3 = 0$ plane, due to symmetry in geometry and loading and $v_3 = 0$ on pore surfaces from no-slip condition
3. v_3 is maximum at pore openings ($x_3 = \pm l_o$)
4. v_1 is symmetric in x_2 and anti-symmetric in x_1
5. v_1 is zero in $x_1 = 0$ plane due to symmetry condition
6. v_2 is symmetric in x_1 and anti-symmetric in x_2
7. v_2 is zero in $x_2 = 0$ plane due to symmetry condition

3.3. Weighted residual method

Parameters A, B, C in Eqs. (33a)–(33c) are obtained by satisfying the continuity equation in weighted integral sense and D, E, F are obtained by satisfying Navier–Stokes equations in weighted integral sense. Weight function for each equation is obtained by taking derivative of the differential equation to be satisfied with respect to the constant to be determined, as per the least square method. The volume flow rate in the pore at a distance x_3 from the center of the pore, can be written as

$$\int_{\Gamma} v_3 dx_1 dx_2 = -r_o^2 \pi x_3 \left(\dot{\bar{\lambda}}_1^{(f)} \bar{\lambda}_2^{(f)} + \dot{\bar{\lambda}}_2^{(f)} \bar{\lambda}_1^{(f)} \right) \quad (34)$$

where Γ is the pore cross-section. From Eqs. (33c) and (34), thus

$$C = -2l_o \left(\frac{\dot{\bar{\lambda}}_1^{(f)}}{\bar{\lambda}_1^{(f)}} + \frac{\dot{\bar{\lambda}}_2^{(f)}}{\bar{\lambda}_2^{(f)}} \right) \quad (35)$$

A and B can be estimated from least square method as (details are given in [Appendix](#))

$$A = -\frac{C \bar{\lambda}_1^{(f)}}{4l_o \dot{\bar{\lambda}}_1^{(f)}} \quad \text{and} \quad B = -\frac{C \bar{\lambda}_2^{(f)}}{4l_o \dot{\bar{\lambda}}_2^{(f)}} \quad (36)$$

Using Eqs. (33a)–(33d) in the Navier–Stokes equations, expressions for the coefficients D, E, F in pressure term are obtained as

$$D = \frac{-\alpha A_1 - 12\alpha^2 - 12\alpha\beta + \alpha S}{2(\alpha A_2 + \beta A_3 + 3\alpha^2 + 2\alpha\beta + 3\beta^2)} \quad (37)$$

$$E = \frac{-\beta A_1 - 12\beta^2 - 12\alpha\beta + \beta S}{2(\alpha A_2 + \beta A_3 + 3\alpha^2 + 2\alpha\beta + 3\beta^2)} \quad (38)$$

$$F = \frac{A_1 + 12\alpha + 12\beta + S}{48} \quad (39)$$

Derivations for D , E , F and the definitions of α , β , S , A_1 , A_2 and A_3 are given in [Appendix](#).

3.4. Viscous dissipation in the fluid

Cauchy stress components σ_{ij} at any point in the fluid domain are given by

$$\sigma_{ij} = -p\delta_{ij} + 2\mu\mathbf{D}_{ij} \quad (i, j = 1, 2, 3) \quad (40)$$

where p is the hydrostatic pressure, μ is the fluid viscosity and δ is kronecker delta. Rate of stretch tensor $\mathbf{D}$ is given by

$$\mathbf{D} = \frac{1}{2} \left[\frac{\partial \mathbf{v}}{\partial \mathbf{x}} + \left(\frac{\partial \mathbf{v}}{\partial \mathbf{x}} \right)^T \right] \quad (41)$$

Thus, the local dissipation density in the fluid is given by

$$\dot{W}^{(f)} = \boldsymbol{\sigma} : \mathbf{D} = 2\mu\mathbf{D}_{ij}\mathbf{D}_{ij} \quad (42)$$

where

$$\mathbf{D}_{ij}\mathbf{D}_{ij} = \left(\frac{\partial v_1}{\partial x_1} \right)^2 + \left(\frac{\partial v_2}{\partial x_2} \right)^2 + \left(\frac{\partial v_3}{\partial x_3} \right)^2 + \frac{1}{2} \left[\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1} \right)^2 + \left(\frac{\partial v_1}{\partial x_3} + \frac{\partial v_3}{\partial x_1} \right)^2 + \left(\frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_2} \right)^2 \right]$$

Since,

$$\frac{\partial v_1}{\partial x_3} = \frac{\partial v_2}{\partial x_3} = 0$$

the expression for local viscous dissipation density becomes

$$\dot{W}^{(f)} = 2\mu \left(\left(\frac{\partial v_1}{\partial x_1} \right)^2 + \left(\frac{\partial v_2}{\partial x_2} \right)^2 + \left(\frac{\partial v_3}{\partial x_3} \right)^2 + \frac{1}{2} \left[\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1} \right)^2 + \left(\frac{\partial v_3}{\partial x_1} \right)^2 + \left(\frac{\partial v_3}{\partial x_2} \right)^2 \right] \right) \quad (43)$$

Substituting the expressions for velocities from Eq. (33a)–(33c) in Eq. (43) and integrating over pore volume, average viscous dissipation density $\dot{W}_{VD}$ in the composite can be obtained as

$$\dot{W}_{VD} = \frac{\mu}{2a_o^2 l_o} \left[2r_o^2 \pi l_o \left(\frac{(A^2 + 2)\bar{\lambda}_1^{-(f)}\bar{\lambda}_2^{-(f)}}{\bar{\lambda}_1^{(f)}} + \frac{(B^2 + 2)\bar{\lambda}_2^{-(f)}\bar{\lambda}_1^{-(f)}}{\bar{\lambda}_2^{(f)}} + \frac{2C^2\bar{\lambda}_1^{-(f)}\bar{\lambda}_2^{-(f)}}{3l^2} + \frac{1}{3} \left(\frac{A\bar{\lambda}_1^{-(f)} + B\bar{\lambda}_2^{-(f)}}{\bar{\lambda}_2^{(f)}\bar{\lambda}_1^{(f)}} \right)^2 \right) \right] + \frac{2\mu C^2 \pi l}{6a_o^2 l_o} \left(\frac{\bar{\lambda}_1^{-(f)}}{\bar{\lambda}_2^{(f)}} + \frac{\bar{\lambda}_2^{-(f)}}{\bar{\lambda}_1^{(f)}} \right) \quad (44)$$

4. Uniaxial plane strain loading

In this section, uniaxial plane strain loading of porous elastomers with mono-dispersed cylindrical voids fully filled with a Newtonian fluid, is considered. The applied macroscopic deformation is characterized by the macroscopic principal stretches $\bar{\lambda}_1 = 1$, $\bar{\lambda}_2 = \bar{\lambda}$ and $\bar{\lambda}_3 = 1$.

4.1. Macroscopic strain energy density $\bar{W}_{SD}$

From Eq. (20), under uniaxial plane strain compression, the macroscopic (average) strain energy density $\bar{W}_{SD}$ (stored energy in fluid-filled elastomer for unit initial volume of fluid-filled elastomer) is given by

$$\bar{W}_{SD}(\bar{\mathbf{F}}) = \frac{G}{2} \frac{(\bar{\lambda}^2 + 1)(\bar{\lambda} - 1)}{2\bar{\lambda}} \log \left(\frac{f_o + \bar{\lambda} - 1}{f_o \bar{\lambda}} \right) + G(1 - f_o) \left(\frac{\bar{\lambda}^2 - 2\bar{\lambda} + 1}{2\bar{\lambda}} \right) \quad (45)$$

Here,

$$\left[\bar{\mathbf{F}} \right] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \bar{\lambda} & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (46)$$

By differentiating Eq. (45) with respect to time, the rate of change of macroscopic strain energy density, for plane strain compression, can be obtained as

$$\dot{\bar{W}}_{SD}(\bar{\mathbf{F}}) = \frac{G\dot{\bar{\lambda}}}{2\bar{\lambda}^2} \left[(1 - f_o) \left((\bar{\lambda}^2 - 1) + \frac{(\bar{\lambda}^2 + 1)(\bar{\lambda} - 1)}{2(f_o + \bar{\lambda} - 1)} \right) + \log \left(\frac{f_o + \bar{\lambda} - 1}{f_o \bar{\lambda}} \right) \frac{2\bar{\lambda}^3 - \bar{\lambda}^2 + 1}{2} \right] \quad (47)$$

4.2. Macroscopic viscous dissipation density $\dot{\bar{W}}_{VD}$

In uniaxial plane strain compression, $\bar{J} = \bar{\lambda}_2 = \bar{\lambda}$, $\dot{\bar{\lambda}}_2 = \dot{\bar{\lambda}}$ and $\dot{\bar{\lambda}}_1 = 0$. Thus, from Eqs. (30) and (31)

$$\begin{aligned}\bar{\lambda}_1^{(f)} &= \sqrt{\frac{(f_o + \bar{\lambda} - 1)}{f_o \bar{\lambda}}}, & \bar{\lambda}_2^{(f)} &= \bar{\lambda} \sqrt{\frac{(f_o + \bar{\lambda} - 1)}{f_o \bar{\lambda}}}, \\ \dot{\bar{\lambda}}_1^{(f)} &= \frac{(1 - f_o) \dot{\bar{\lambda}}}{2 f_o^{1/2} \bar{\lambda}^{-3/2} \sqrt{f_o + \bar{\lambda} - 1}}, & \dot{\bar{\lambda}}_2^{(f)} &= \frac{\dot{\bar{\lambda}}(f_o + 2\bar{\lambda} - 1)}{2 f_o^{1/2} \bar{\lambda}^{-1/2} \sqrt{f_o + \bar{\lambda} - 1}}.\end{aligned}\quad (48)$$

Substituting Eq. (48) in Eq. (44), for $r_o \ll 2 l_o$, macroscopic viscous dissipation density (viscous dissipation per unit initial volume of the fluid-filled porous elastomer) can be estimated as

$$\dot{\bar{W}}_{VD} = \frac{4\mu\pi\dot{\bar{\lambda}}^2}{3(f_o + \bar{\lambda} - 1)^2} \left(\frac{1}{\bar{\lambda}} + \bar{\lambda} \right) \left(\frac{l_o^2}{a_o^2} \right). \quad (49)$$

4.3. Small strain estimates

4.3.1. Macroscopic strain energy density

In the limit of small deformations ($\bar{\epsilon} = \bar{\lambda} - 1 \rightarrow 0$), the macroscopic strain energy density function (20) and its rate of change can be estimated as

$$\bar{W}_{SD}(\bar{\mathbf{F}}) = \frac{G}{2} \frac{(1 - f_o^2)}{f_o} \bar{\epsilon}^2 \quad \text{and} \quad \dot{\bar{W}}_{SD}(\bar{\mathbf{F}}) = G \frac{(1 - f_o^2)}{f_o} \bar{\epsilon} \dot{\bar{\epsilon}} \quad (50)$$

In the limit of small deformations and small porosities ($f_o, \bar{\epsilon} \rightarrow 0$), the above equation can be further simplified as

$$\bar{W}_{SD}(\bar{\mathbf{F}}) = \frac{G}{2} \frac{\bar{\epsilon}^2}{f_o} \quad \text{and} \quad \dot{\bar{W}}_{SD}(\bar{\mathbf{F}}) = G \frac{\bar{\epsilon} \dot{\bar{\epsilon}}}{f_o} \quad (51)$$

4.3.2. Macroscopic viscous dissipation density

In the limit of small deformations, for finite porosity, macroscopic viscous dissipation density of fluid-filled porous elastomers under plane strain compression can be shown as

$$\dot{\bar{W}}_{VD} = \frac{8\mu\pi l_o^2 \dot{\bar{\epsilon}}^2}{3 f_o^2 a_o^2}. \quad (52)$$

In the limit of small porosities and small deformations, macroscopic viscous dissipation density is given by

$$\dot{\bar{W}}_{VD} = \frac{8\mu\pi l_o^2 \dot{\bar{\epsilon}}^2}{3(f_o + \bar{\epsilon})^2 a_o^2}. \quad (53)$$

4.4. Equivalent spring stiffness and damping coefficient

Rate of external work done per unit volume of fluid-filled elastomer can be written as

$$\dot{\bar{W}} = \dot{\bar{W}}_{SD} + \dot{\bar{W}}_{VD} \quad (54)$$

where, for small deformations and uniaxial plane strain loading

$$\dot{\bar{W}} = \bar{\sigma} \dot{\bar{\epsilon}} \quad (55)$$

Substituting the expressions for $\dot{\bar{W}}$, $\dot{\bar{W}}_{SD}$, $\dot{\bar{W}}_{VD}$ from Eq. (55), Eq. (50) and (52) in Eq. (54), for finite porosities and small strains, it can be shown that

$$\bar{\sigma} = G \frac{(1 - f_o^2)}{f_o} \bar{\epsilon} + \frac{8\mu\pi l_o^2 \dot{\bar{\epsilon}}}{3 f_o^2 a_o^2} \quad (56)$$

Substituting $\bar{\epsilon} = \epsilon_0 e^{i\omega t}$, $\bar{\sigma} = \sigma_0 e^{i(\omega t + \delta)}$ in the above expression for harmonic loading, the complex modulus of the composite material can be obtained as

$$E^* = \frac{\sigma}{\epsilon} = E' + iE'' = G \frac{(1 - f_o^2)}{f_o} + i\omega \frac{8\mu\pi l_o^2}{3 f_o^2 a_o^2} \quad (57)$$

From the above equation, the storage modulus E' and the loss modulus E'' can be obtained as

$$E' = G \frac{(1 - f_o^2)}{f_o}, \quad E'' = \omega \frac{8\mu\pi l_o^2}{3 f_o^2 a_o^2} = \frac{8\mu\omega}{3 f_o} \left(\frac{l_o}{r_o} \right)^2. \quad (58)$$

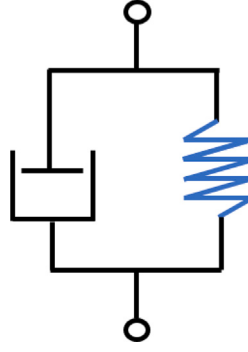


Fig. 5. Kelvin–Voigt model for composite.

It is clear from above expressions that the factors affecting the storage modulus are the initial porosity and the shear modulus of the elastomer. On the other hand, the loss modulus is affected by the frequency of applied loading, the initial porosity, fluid viscosity, and the ratio of pore radius to pore length. The damping ratio is estimated as

$$\tan \delta = \frac{E''}{E'} = \frac{8}{3} \frac{\mu \omega l_o^2}{G r_o^2 (1 - f_o^2)}. \quad (59)$$

An important observation from Eqs. (58) and (59) is that the loss modulus and the damping factor are inversely proportional to the square of pore aspect ratio ($r_o/2l_o$). Considering Kelvin–Voigt model (see Fig. 5) for the fluid-filled elastomer subjected to plane strain compression (see Fig. 3), the equivalent spring stiffness K and damping constant C can be obtained (for finite porosity) as (see Fig. 5)

$$K = 2l_o E' = 2G l_o \frac{(1 - f_o^2)}{f_o} \quad \text{and} \quad (60)$$

$$C = \frac{2l_o E''}{\omega} = \frac{16 \mu l_o^3}{3 f_o r_o^2}. \quad (61)$$

5. Validation of analytical estimates

5.1. Velocity and pressure fields in the fluid domain

Figs. 6–9 show a comparison of the numerical results and theoretical estimates for fluid pressure and velocity components v_1, v_2, v_3 , from deformation induced flow for the fluid-filled elastomer considered in this work. Here, $G = 369.4$ kPa for the incompressible Neo-Hookean matrix, fluid viscosity $\mu = 1$ Pa s and pore aspect ratio $2l_o/r_o = 20$. Plane strain loading was considered such that the length of the pore remains unchanged during the compression. Pressure at pore ends is taken to be zero, i.e. $p_o = 0$. Simulation results shown are for “Normal mesh” (see Table 1 for details of the mesh). Results obtained with “Normal” and “Fine” meshes are nearly identical. “Normal” mesh is chosen for faster computation of results.

An excellent agreement is seen between numerical results and analytical estimates at all stretches considered. Pressure variation in the fluid along the pore axis (see Fig. 6) is parabolic with the maximum pressure occurring at the pore center. From Fig. 7, it can be seen that v_3 has a linear variation along the pore axis. v_3 is maximum at the pore openings and is zero at pore boundary surface (due to no-slip boundary condition) and at pore center (due to symmetry conditions). Further, the maximum value of v_3 increases with compression. At all compression levels, the match between the numerical results and theoretical estimates is excellent. From Fig. 8, it can be observed that even when compression occurs in x_2 direction the pore wall has some movement in the x_1 direction and v_1 is non-zero at the walls. The difference in numerical and theoretical results is small (see Table 2 for goodness of fit). The velocity component v_2 (see Fig. 9) remains unchanged with compression in numerical and theoretical results. Thus, the proposed pressure and velocity estimates for fluid are very accurate, at least for Newtonian fluids.

Table 2 shows the (R^2 value) goodness of fit between analytical and numerical results for pressure and velocities. R^2 value for pressure is 1 during compression. Therefore, numerical and analytical results match perfectly. For v_1, v_2 and v_3 they are in the range 0.988 to 0.999 which indicates a very good match between analytical estimates and numerical results. Figs. 10–12 present comparative contour plots for v_1, v_2, v_3 in the x_1x_2 plane (cross-sectional plane) located at $x_3 = l_o/2$ for a macroscopic stretch of $\bar{\lambda} = 0.98$. For v_1 and v_3 top halves are numerical results and bottom halves are analytical estimates, whereas for v_2 left half is the numerical result and the right half is the analytical estimate. Contours plots confirm that the match between numerical and analytical results for v_1, v_2 and v_3 is excellent at all locations.

Fig. 13 shows the comparative contour plot for pressure variation along pore length in the x_1x_3 plane ($x_2 = 0$) for a macroscopic stretch of $\bar{\lambda} = 0.98$. Pressure variation along the pore length is very well predicted by the proposed analytical function. As can be seen, the color bands from numerical results match very well with the analytical predictions.

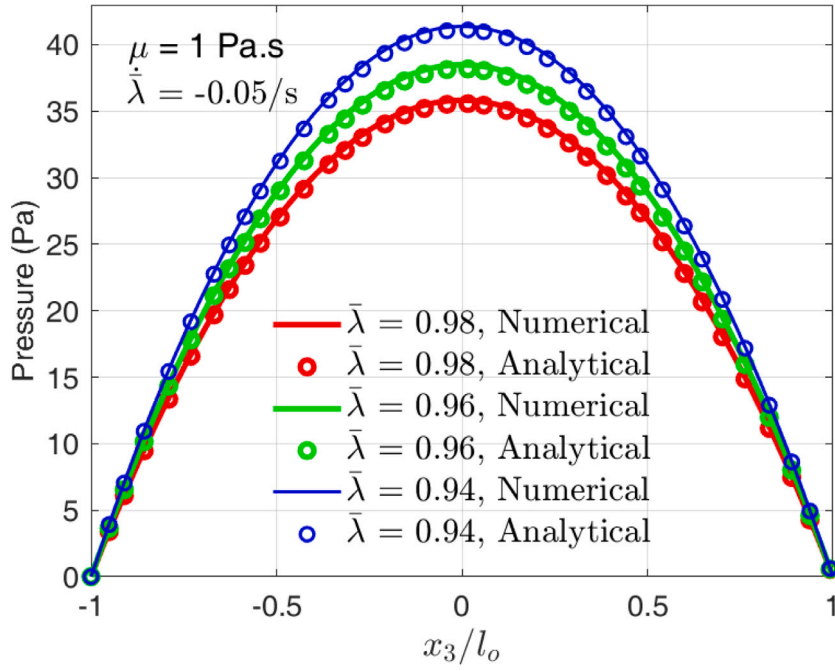


Fig. 6. Comparison of numerical and theoretical results for fluid pressure along pore axis ($x_1 = 0$, $x_2 = 0$). Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\lambda} = -0.05/s$, $p_o = 0$, macroscopic stretch $\bar{\lambda} = 0.98, 0.96$ and 0.94 ($\bar{\epsilon} = 2\%, 4\%$ and 6%).

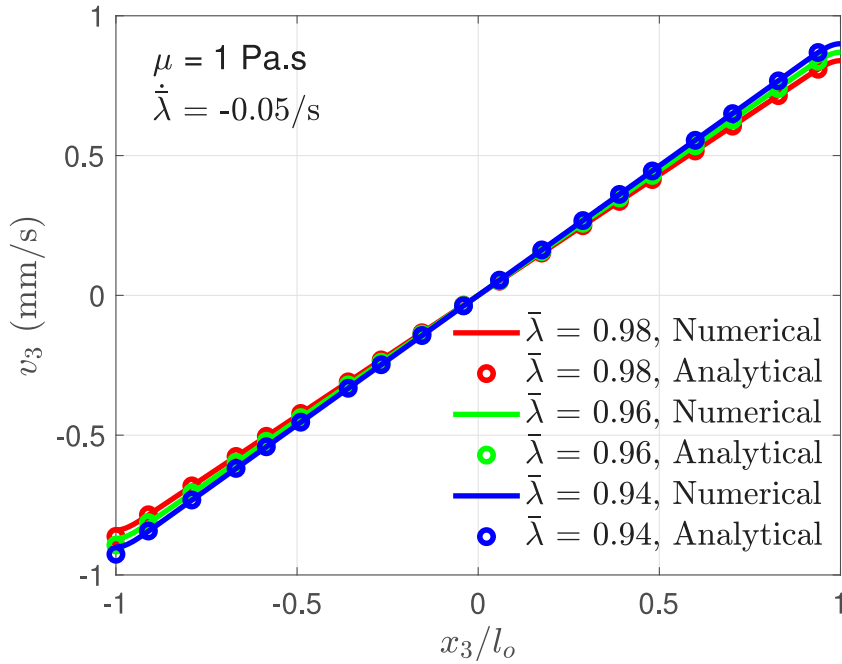


Fig. 7. Comparison of numerical and theoretical results for fluid velocity v_3 along pore axis ($x_1 = 0$, $x_2 = 0$). Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\lambda} = -0.05/s$, macroscopic stretch $\bar{\lambda} = 0.98, 0.96$ and 0.94 .

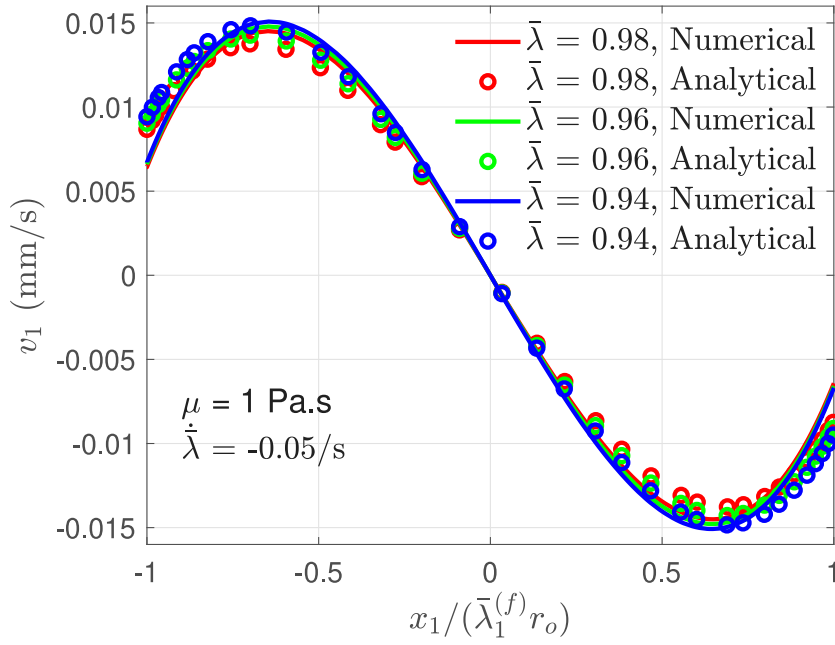


Fig. 8. Comparison of numerical and theoretical results for v_1 along x_1 -direction at $x_2 = 0$ and $x_3 = l_o/2$. Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\lambda} = -0.05/\text{s}$, macroscopic stretch $\lambda = 0.98, 0.96$ and 0.94 .

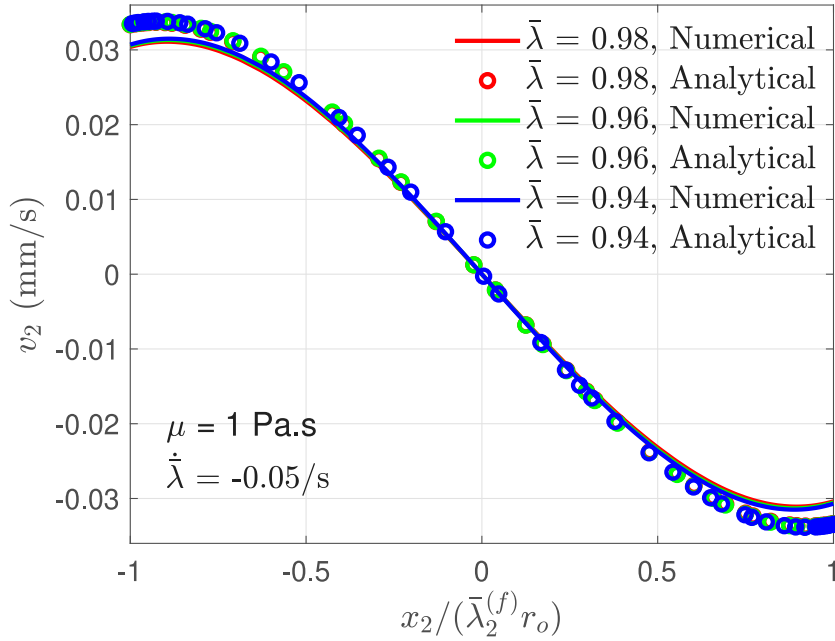


Fig. 9. Comparison of numerical and theoretical results for v_2 along x_2 -direction at $x_1 = 0$ and $x_3 = l_o/2$. Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\lambda} = -0.05/\text{s}$, macroscopic stretch $\lambda = 0.98, 0.96$ and 0.94 .

5.2. Validation of estimates for macroscopic strain energy and viscous dissipation densities

Fig. 14 presents comparison of numerical results and theoretical estimates for macroscopic strain energy density of the fluid-filled elastomer subjected to uniaxial plane strain compression with $\bar{\lambda}_1 = 1, \bar{\lambda}_2 = \bar{\lambda}$ and $\bar{\lambda}_3 = 1$. The strain energy density estimates have

Table 2
Goodness of fit between numerical and analytical results.

Variable	R ² value		
	$\bar{\lambda} = 0.98$	$\bar{\lambda} = 0.96$	$\bar{\lambda} = 0.94$
Pressure	1	1	1
v_1	0.989648	0.989568	0.988834
v_2	0.999869	0.999840	0.999832
v_3	0.999973	0.999973	0.999968

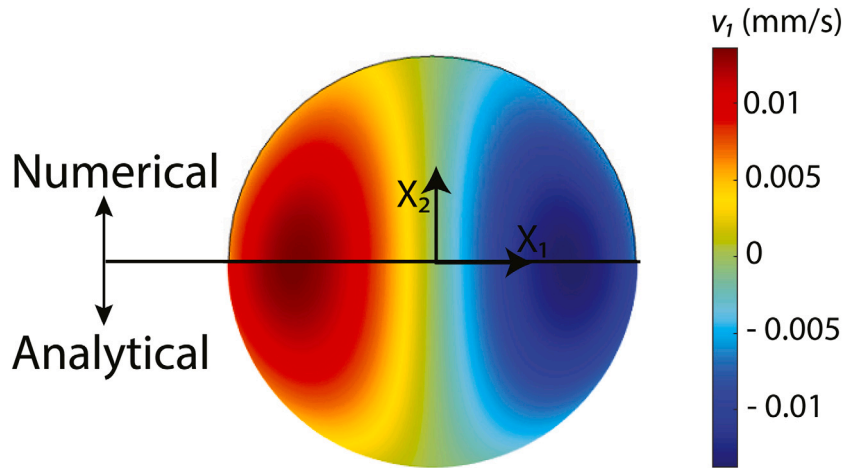


Fig. 10. Comparison contour plots for velocity component v_1 in the x_1x_2 plane at $x_3 = l_0/2$. Top half shows the simulation result while bottom half shows the analytical result. Note that v_1 is symmetric about x_1x_3 plane. Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\bar{\lambda}} = -0.05/s$, macroscopic stretch $\bar{\lambda} = 0.98$.

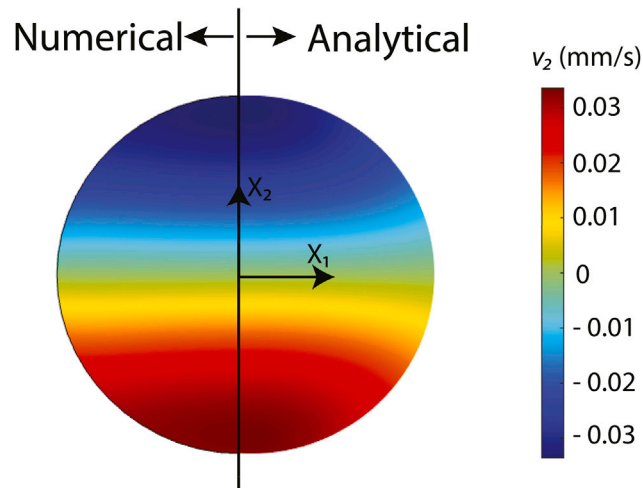


Fig. 11. Comparison contour plots for velocity component v_2 in the x_1x_2 plane at $x_3 = l_0/2$. Left half shows the simulation result while right half shows the analytical result. Note that v_2 is symmetric about x_2x_3 plane. Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\bar{\lambda}} = -0.05/s$, macroscopic stretch $\bar{\lambda} = 0.98$.

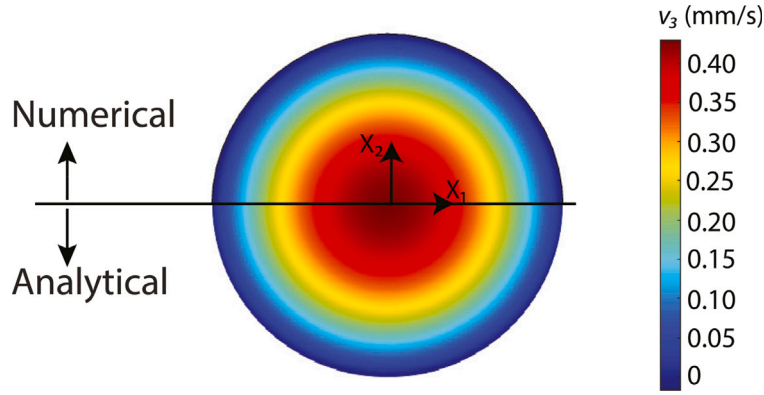


Fig. 12. Comparison contour plots for velocity component v_3 in the x_1x_2 plane at $x_3 = l_0/2$. Top half shows the simulation result while bottom half shows the analytical result. Note that v_3 is symmetric about x_1x_3 plane. Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\lambda} = -0.05/s$, macroscopic stretch $\bar{\lambda} = 0.98$.

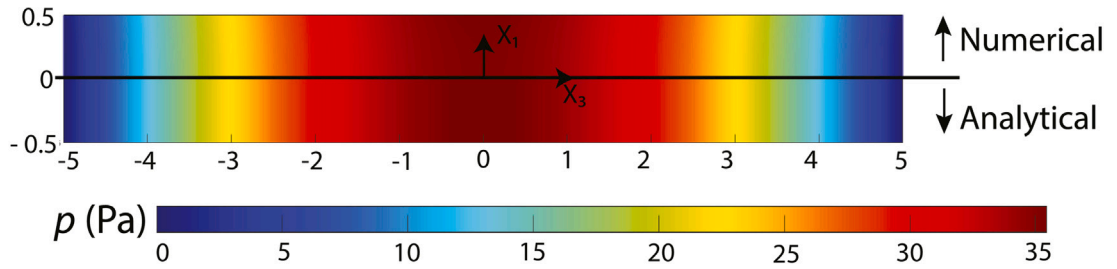


Fig. 13. Comparison contour plot for pressure p in the x_1x_3 plane located at $x_2 = 0$. Top half shows the simulation result while bottom half shows the analytical result. Results are shown for Newtonian fluid with viscosity $\mu = 1$ Pa s, initial pore radius $r_o = 0.5$ mm, loading rate $\dot{\lambda} = -0.05/s$, macroscopic stretch $\bar{\lambda} = 0.98$.

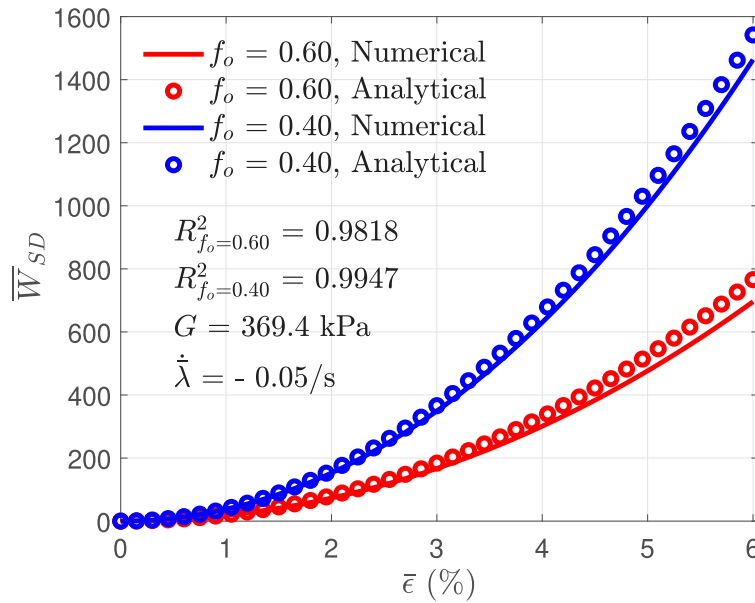


Fig. 14. Comparison of numerical results and theoretical estimates for macroscopic strain energy density of fluid-filled elastomers for uniaxial compression ($\bar{\lambda}_1 = 1, \bar{\lambda}_2 = 1 - \bar{\epsilon}, \bar{\lambda}_3 = 1$). Results are shown for an incompressible Neo-Hookean elastomer with initial void fraction $f_o = 0.4$ and 0.6 . Solid lines are used for numerical results and circular markers are used for analytical estimates.

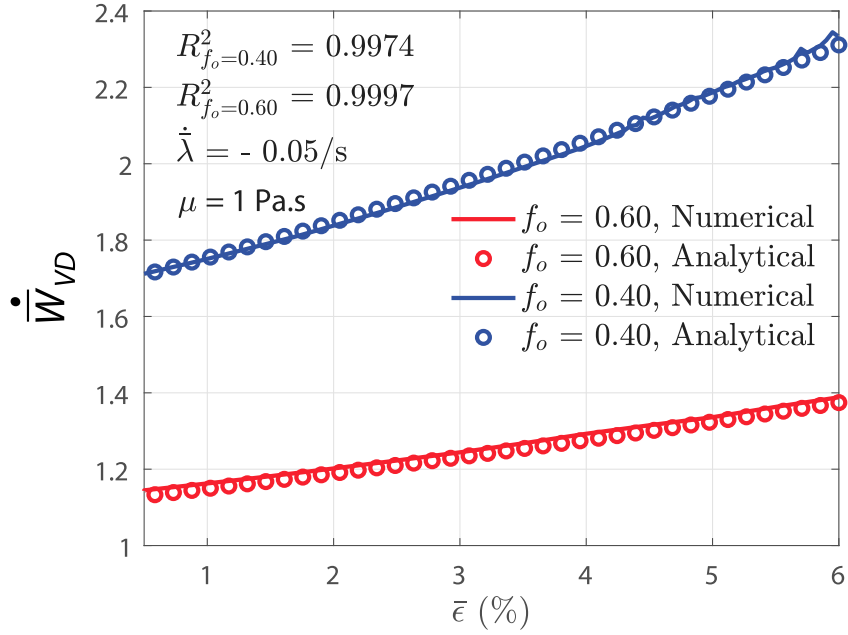


Fig. 15. Comparison of numerical and theoretical estimates for macroscopic viscous dissipation density in the fluid-filled elastomeric composite subjected to uniaxial compression ($\bar{\lambda}_1 = 1, \bar{\lambda}_2 = 1 - \bar{\epsilon}, \bar{\lambda}_3 = 1$). Results are shown for a Newtonian fluid with viscosity $\mu = 1 \text{ Pa}\cdot\text{s}$ and applied macroscopic loading rate $\dot{\lambda} = -0.05/s$. Solid lines are used for numerical results and circular markers are used for analytical estimates.

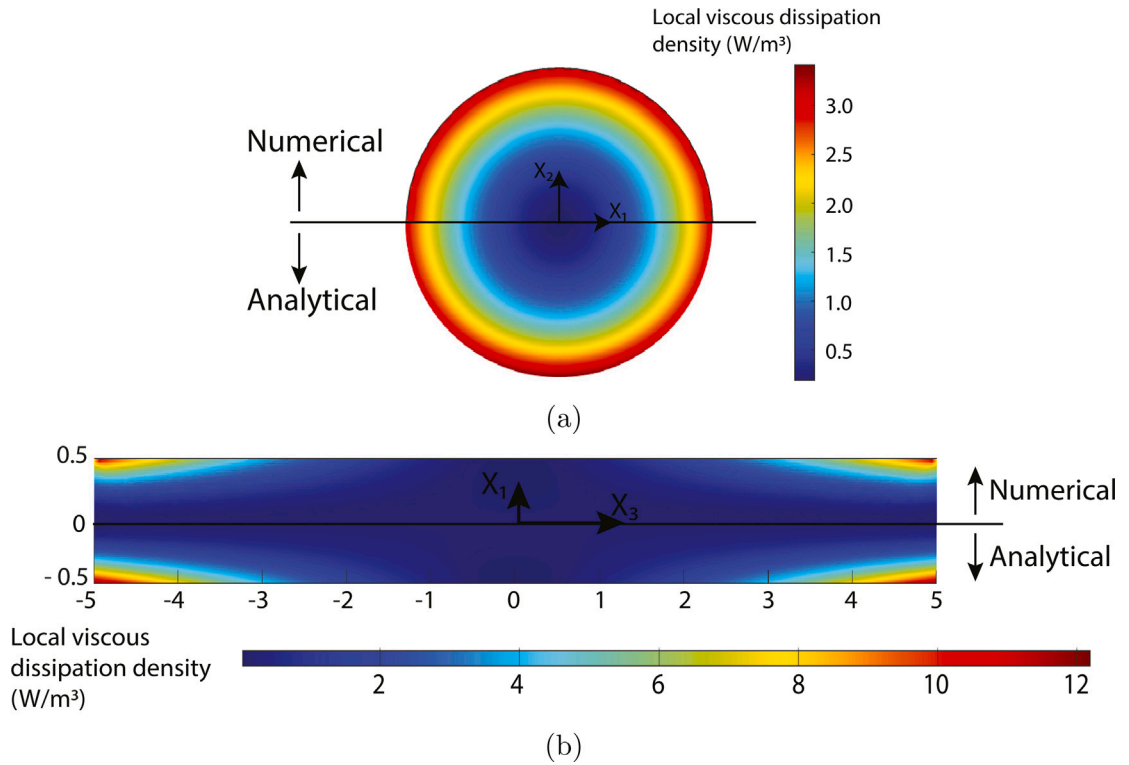


Fig. 16. Comparison contour plot for local viscous dissipation density in (a) the x_1x_2 plane located at $x_3 = l_o/2$ and (b) in the x_1x_3 plane located at $x_2 = 0$. Top half shows simulation results while bottom half shows analytical results. Results are shown for Newtonian fluid with viscosity $\mu = 1 \text{ Pa}\cdot\text{s}$, initial pore radius $r_o = 0.5 \text{ mm}$, loading rate $\dot{\lambda} = -0.05/s$, macroscopic stretch $\bar{\lambda} = 0.98$.

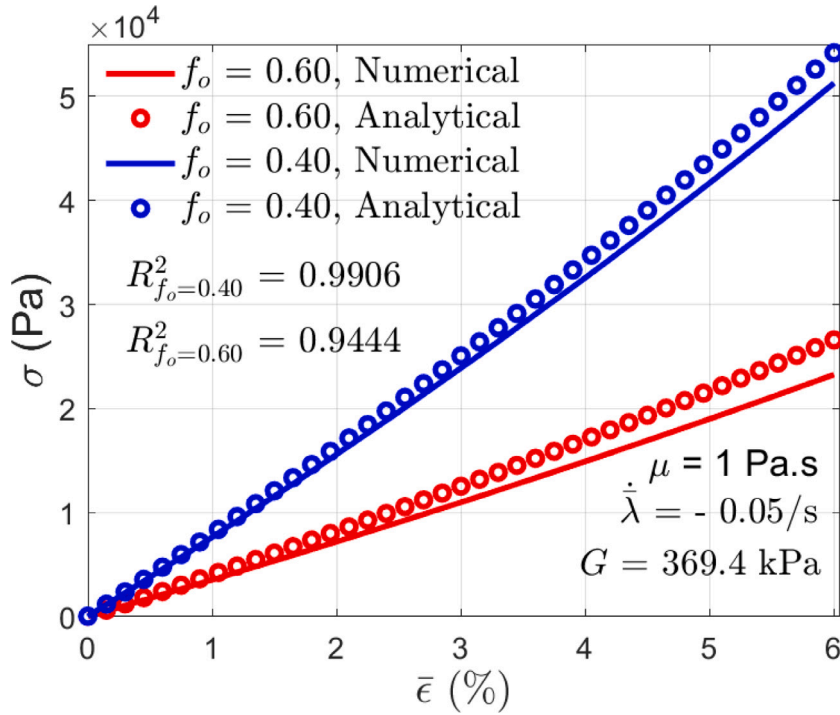


Fig. 17. Stress-strain curve for the fluid-filled elastomeric composite subjected to uniaxial compression ($\bar{\lambda}_1 = 1, \bar{\lambda}_2 = \bar{\lambda}, \bar{\lambda}_3 = 1$). Results are shown for a Neo-Hookean elastomer with shear modulus $G = 369.4$ kPa and Newtonian fluid with viscosity $\mu = 1$ Pa s for applied macroscopic loading rate $\dot{\bar{\lambda}} = -0.05/s$ and initial porosity $f_o = 40\%$ and 60% . The results are shown as a function of percentage strain $\bar{\epsilon} = 1 - \bar{\lambda}$. Solid lines are for numerical results and circular markers are for analytical results.

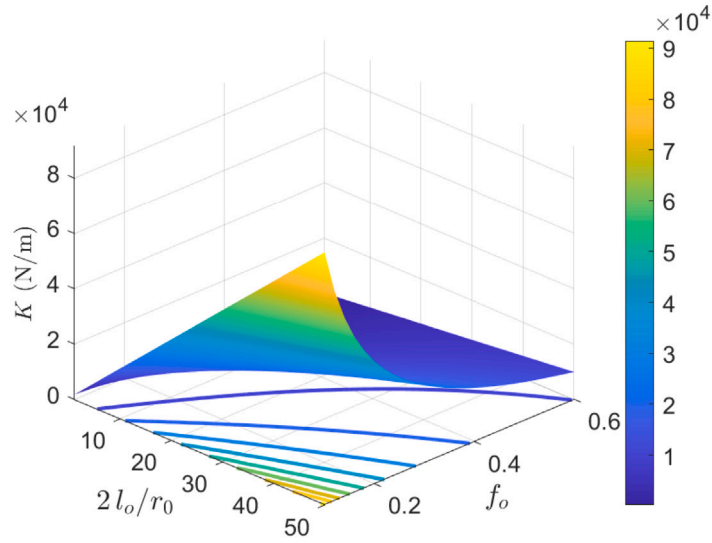
been plotted for initial porosities $f_o = 40\%$ and 60% , as a function of the percentage compressive strain $\bar{\epsilon} = 1 - \bar{\lambda}$. Strain energy density from analytical estimates is seen to be higher than that from numerical results. The analytical estimates are slightly higher than numerical results as a kinematically admissible displacement field is considered for the elastomer that does not necessarily minimize the potential energy. Thus, the analytical estimate for macroscopic strain energy density is an upper bound. The estimate is quite accurate for small strains.

Fig. 15 presents the comparison of macroscopic viscous dissipation density in the fluid when the composite is subjected to uniaxial compression. Results are shown for two different initial porosities as a function of the percentage strain. Viscous dissipation density is significantly higher for the lower porosity case. It is seen that agreement between analytical and numerical results is excellent. In the present study, pore aspect ratio $2l_o/r_o = 20$. Fig. 16 presents a comparative contour plots for viscous dissipation in the x_1x_2 plane (cross-sectional plane) at $x_3 = l_o/2$ and in the x_1x_3 plane located at $x_2 = 0$ for a macroscopic stretch of $\bar{\lambda} = 0.98$. A good match between the numerical and analytical results is observed at all locations. Fig. 16 shows a comparative contour plot for local viscous dissipation density in x_1x_2 and x_1x_3 plane.

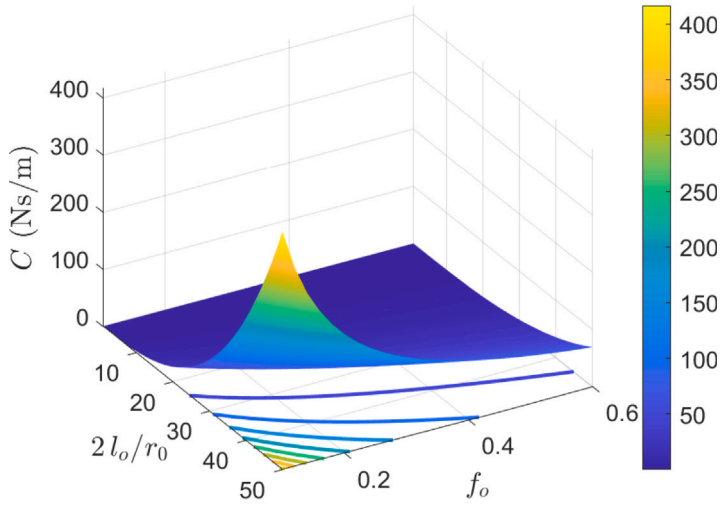
Fig. 17 shows the stress-strain curve for the composite. The stress-strain curve is linear for small deformations. Since the response for fluid phase is sensitive to loading rate, due to the small loading rate considered, the role of fluid is suppressed. It can be seen that equivalent modulus (slope of stress-strain curve) is inversely proportional to porosity. Match between analytical and numerical results is good at small deformations. The match reduces with increasing deformation.

5.3. Equivalent spring stiffness and damping coefficient

Fig. 18 shows contour plots for the equivalent spring stiffness K and damping coefficient C of the composite material when it is subjected to uniaxial compression, as a function of initial porosity f_o and initial pore aspect ratio $2l_o/r_o$. As seen from the Eq. (60), K is linearly dependent on initial pore aspect ratio and in a non-linear way on initial porosity. For small porosities the spring stiffness is inversely proportional to initial porosity. Therefore, the spring stiffness increases sharply as initial porosity becomes small. It can be seen from Fig. 18(b) that for a given pore radius as initial pore fraction decreases and initial pore aspect ratio increases the damping constant increases.



(a)



(b)

Fig. 18. Contour plot showing (a) equivalent spring stiffness and (b) equivalent damping coefficient of fluid-filled elastomeric composites. Results are plotted as a function of pore aspect ratio ($2l_o/r_o$) and initial pore fraction f_o for $r_o = 0.5$ mm.

Concluding remarks

In the present work, analytical estimates are developed for fluid-filled elastomers for average deformation gradient, average strain energy density of solid phase, deformation of pores, velocities and pressure fields in pores, viscous dissipation in fluid and the overall response of fluid-filled elastomer. The macroscopic strain energy density is seen to be a function of matrix shear modulus, initial porosity and applied macroscopic deformation. Predictions of macroscopic strain energy density function were compared with the results from unit-cell numerical simulations. The agreement between the analytical estimates and numerical results was good for the considered loadings and initial porosities.

Analytical estimates for pressure and velocity field are derived for deformation induced fluid flow, using a weighted residual (least square) method. Pressure and velocity functions are assumed such that they satisfied the boundary conditions exactly. The unknown constants in the proposed pressure and velocity functions are obtained by enforcing Navier–Stokes and continuity equations in weighted integral sense. The developed framework was applied to an incompressible Newtonian fluid. The analytical predictions

were compared with results from numerical simulations to validate the derived analytical estimates for pressure, velocity components and viscous dissipation in the fluid. Excellent agreement is seen between the numerical and analytical results for different initial porosities. The frameworks for elastomeric matrix deformation and fluid flow were used to estimate the overall response of the fluid-filled composite. The analytical predictions for the overall response matched well with the results obtained from numerical simulations particularly for small deformations.

Finally, the storage and loss moduli are obtained for the fluid-filled elastomeric composites. Storage modulus is seen to be a function of shear modulus of elastomer and initial porosity. Loss moduli on the other hand is a function of fluid viscosity, pore length, initial porosity, pore radius and frequency of loading. The loss modulus being directly proportional to applied load frequency is a unique property of the composite. Loss and storage moduli are used to estimate the equivalent spring stiffness and equivalent damping coefficient of the composite.

The present work explores an exciting class of materials, i.e. fluid-filled elastomers. These materials can be used to replace conventional spring-damper systems. The main factors governing the properties of the material are initial porosity, elastomer properties and fluid properties and pore aspect ratio. Since these factors can be easily customized, it is possible to custom develop a material as per requirements in an application.

CRedit authorship contribution statement

Vivek Singh: Methodology, Software, Validation, Formal analysis, Visualization, Investigation, Writing – original draft & editing.
Vikranth Racherla: Conceptualization, Methodology, Software, Validation, Formal analysis, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Here a detailed mathematical derivation is given for various constants in velocity and pressure functions. As discussed in the main text, we use least square method for obtaining the constants.

A.1. Least square method

First we solve for constants A and B in the proposed velocity functions. Proposed velocity functions are substituted in the continuity equation to get the residual term. We now differentiate the residual with respect to A and B to get 2 weight functions. Now the product of residual and weight function is integrated over the fluid domain to get 2 linear equations in A and B . Solving the linear equations we get the expressions for A and B . For D , E and F proposed velocity and pressure functions are substituted in Navier–Stokes equations to get the residual terms for the 3-directions. While obtaining the weight functions for Navier–Stokes equations we differentiate the residual term for a particular direction is differentiated with respect to the coefficient for that direction. The expression for C was obtained in the main text itself. Here we give the derivations for A , B , D , E and F .

A.2. Constants in velocity functions (A , B)

For this case, the residual is obtained from the continuity equation, and the expressions for v_1 , v_2 and v_3 are taken from (33a)–(33c), thus

$$R_1 = \nabla \cdot \mathbf{v}$$

The derivative of the residual R_1 with respect to A , B give

$$\frac{\partial R_1}{\partial A} = \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}} \left(1 - \frac{3x_1^2}{\lambda_1^{(f)2} r^2} - \frac{x_2^2}{\lambda_2^{(f)2} r^2} \right) \quad \text{and} \quad \frac{\partial R_1}{\partial B} = \frac{\dot{\lambda}_2^{(f)}}{\lambda_2^{(f)}} \left(1 - \frac{x_1^2}{\lambda_1^{(f)2} r^2} - \frac{3x_2^2}{\lambda_2^{(f)2} r^2} \right)$$

Now applying least square method and integrating the product of residual and the derivative of residual with respect to constant, one by one for A and B , separately we obtain

$$\int_{\Omega} R_1 \frac{\partial R_1}{\partial A} d\Omega = 0 \quad \rightarrow \quad B l_o \frac{\dot{\lambda}_2^{(f)}}{\lambda_2^{(f)}} \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}} + (C \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}} + 3 A l_o \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}}) \frac{\dot{\lambda}_2^{(f)}}{\lambda_2^{(f)}} = 0 \quad (62)$$

$$\int_{\Omega} R_1 \frac{\partial R_1}{\partial B} d\Omega = 0 \quad \rightarrow \quad 3 B l_o \frac{\dot{\lambda}_2^{(f)}}{\lambda_2^{(f)}} \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}} + (C \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}} + A l_o \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}}) \frac{\dot{\lambda}_2^{(f)}}{\lambda_2^{(f)}} = 0 \quad (63)$$

Solving (62) and (63) we get the expression for A , B .

$$A = -\frac{C \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}}}{4 l_o \frac{\dot{\lambda}_1^{(f)}}{\lambda_1^{(f)}}} \quad \text{and} \quad B = -\frac{C \frac{\dot{\lambda}_2^{(f)}}{\lambda_2^{(f)}}}{4 l_o \frac{\dot{\lambda}_2^{(f)}}{\lambda_2^{(f)}}}$$

A.3. Constants in pressure function (D , E , F)

For solving the constants in pressure function, the Navier–Stokes equations are enforced. More specifically we take

$$\int_{\Omega} R_2 \frac{\partial R_2}{\partial D} d\Omega = 0$$

with

$$R_2 \equiv \rho \left(\frac{\partial v_1}{\partial t} + v_1 \frac{\partial v_1}{\partial x_1} + v_2 \frac{\partial v_1}{\partial x_2} + v_3 \frac{\partial v_1}{\partial x_3} \right) + \frac{\partial p}{\partial x_1} - \mu \left(\frac{\partial^2 v_1}{\partial x_1^2} + \frac{\partial^2 v_1}{\partial x_2^2} + \frac{\partial^2 v_1}{\partial x_3^2} \right)$$

Solving the least-square equation with residual equation for x_1 -direction momentum equation and the constant term associated with x_1 -direction in pressure term (i.e. D).

$$D F + \frac{5 r^2 \rho \dot{\lambda}_1^{(f)}}{48} \left[2 \dot{\lambda}_1^{(f)} \frac{\partial A}{\partial t} + \frac{2 A \dot{\lambda}_1^{(f)} \dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} + 2 \dot{\lambda}_1^{(f)} (2A - 3) - \left((A^2 + 4A - 12) \dot{\lambda}_1^{(f)} + \frac{A(4 + B) \dot{\lambda}_2^{(f)} \dot{\lambda}_1^{(f)}}{\dot{\lambda}_2^{(f)}} \right) \right] + \frac{5 \mu \dot{\lambda}_1^{(f)} \dot{\lambda}_1^{(f)} A}{4} \left(\frac{3}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right) = 0 \quad (64)$$

Similarly, we take

$$\int_{\Omega} R_3 \frac{\partial R_3}{\partial E} d\Omega = 0$$

with

$$R_3 \equiv \rho \left(\frac{\partial v_2}{\partial t} + v_1 \frac{\partial v_2}{\partial x_1} + v_2 \frac{\partial v_2}{\partial x_2} + v_3 \frac{\partial v_2}{\partial x_3} \right) + \frac{\partial p}{\partial x_2} - \mu \left(\frac{\partial^2 v_2}{\partial x_1^2} + \frac{\partial^2 v_2}{\partial x_2^2} + \frac{\partial^2 v_2}{\partial x_3^2} \right)$$

Solving the least-square equation with residual equation for x_2 -direction momentum equation we get

$$E F + \frac{5 r^2 \rho \dot{\lambda}_2^{(f)}}{48} \left[2 \dot{\lambda}_2^{(f)} \frac{\partial B}{\partial t} + \frac{2 B \dot{\lambda}_2^{(f)} \dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + 2 \dot{\lambda}_2^{(f)} (2B - 3) - \left((B^2 + 4B - 12) \dot{\lambda}_2^{(f)} + \frac{B(4 + A) \dot{\lambda}_1^{(f)} \dot{\lambda}_2^{(f)}}{\dot{\lambda}_1^{(f)}} \right) \right] + \frac{5 \mu \dot{\lambda}_2^{(f)} \dot{\lambda}_2^{(f)} B}{4} \left(\frac{3}{\dot{\lambda}_2^{(f)2}} + \frac{1}{\dot{\lambda}_1^{(f)2}} \right) = 0 \quad (65)$$

Lastly, x_3 -direction momentum equation gives

$$\int_{\Omega} R_4 \frac{\partial R_4}{\partial F} d\Omega = 0$$

with

$$R_4 \equiv \rho \left(\frac{\partial v_3}{\partial t} + v_1 \frac{\partial v_3}{\partial x_1} + v_2 \frac{\partial v_3}{\partial x_2} + v_3 \frac{\partial v_3}{\partial x_3} \right) + \frac{\partial p}{\partial x_3} - \mu \left(\frac{\partial^2 v_3}{\partial x_1^2} + \frac{\partial^2 v_3}{\partial x_2^2} + \frac{\partial^2 v_3}{\partial x_3^2} \right)$$

Solving the least-square equation with residual equation for z -direction momentum equation we get

$$\begin{aligned} & \rho \left[8C^2 + 6C l_o^2 \left(\dot{\lambda}_1^{(f)} \dot{\lambda}_1^{(f)} + \dot{\lambda}_2^{(f)} \dot{\lambda}_2^{(f)} \right) - 4C l_o \left((3 + A) \frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + (3 + B) \frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) + 6 l_o \frac{\partial C}{\partial t} \right] \\ & + D \rho \left[C^2 + C l_o^2 (3 \dot{\lambda}_1^{(f)} \dot{\lambda}_1^{(f)} + \dot{\lambda}_2^{(f)} \dot{\lambda}_2^{(f)}) - \frac{C l_o}{2} \left((12 + 3A) \frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + (4 + B) \frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) + \frac{\partial C}{\partial t} l_o \right] \\ & + E \rho \left[C^2 + C l_o^2 (\dot{\lambda}_1^{(f)} \dot{\lambda}_1^{(f)} + 3 \dot{\lambda}_2^{(f)} \dot{\lambda}_2^{(f)}) - \frac{C l_o}{2} \left((4 + A) \frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + (12 + 3B) \frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) + \frac{\partial C}{\partial t} l_o \right] \\ & + \frac{24 \mu C l_o}{r^2} \left(\frac{1}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right) + \frac{6 \mu C D l_o}{r^2} \left(\frac{1}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right) + \frac{6 \mu C E l_o}{r^2} \left(\frac{1}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right) \\ & - F (3 (D^2 + E^2) + 12 (D + E) + 2DE + 24) = 0 \end{aligned} \quad (66)$$

The constants in the expressions for D , E and F are given below

$$\begin{aligned}
 S &= \sqrt{A_1^2 + 24(\alpha + \beta)A_1 - 144\alpha^2 - 144\beta^2 - 96\alpha A_2 - 96\alpha A_3 + 96\alpha\beta} \\
 \alpha &= \frac{5r^2\rho\dot{\lambda}_1^{(f)}}{48} \left[2\frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_2^{(f)}} \frac{\partial A}{\partial t} + \frac{2A\dot{\lambda}_1^{(f)}\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} + 2\dot{\lambda}_1^{(f)}(2A-3) \right. \\
 &\quad \left. - \left((A^2 + 4A - 12)\dot{\lambda}_1^{(f)} + \frac{A(4+B)\dot{\lambda}_2^{(f)}\dot{\lambda}_1^{(f)}}{\dot{\lambda}_2^{(f)}} \right) \right] + \frac{5\mu\dot{\lambda}_1^{(f)}\dot{\lambda}_{f1}^{(f)}A}{4} \left(\frac{3}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right), \\
 \beta &= \frac{5r^2\rho\dot{\lambda}_2^{(f)}}{48} \left[2\frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_1^{(f)}} \frac{\partial B}{\partial t} + \frac{2B\dot{\lambda}_2^{(f)}\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + 2\dot{\lambda}_2^{(f)}(2B-3) \right. \\
 &\quad \left. - \left((B^2 + 4B - 12)\dot{\lambda}_2^{(f)} + \frac{B(4+A)\dot{\lambda}_1^{(f)}\dot{\lambda}_2^{(f)}}{\dot{\lambda}_1^{(f)}} \right) \right] + \frac{5\mu\dot{\lambda}_2^{(f)}\dot{\lambda}_2^{(f)}A}{4} \left(\frac{3}{\dot{\lambda}_2^{(f)2}} + \frac{1}{\dot{\lambda}_1^{(f)2}} \right), \\
 A_1 &= \rho \left[8C^2 + 6Cl_o^2 \left(\frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + \frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) - 4Cl_o \left((3+A)\frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + (3+B)\frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) + 6l_o \frac{\partial C}{\partial t} \right] \\
 &\quad + \frac{24\mu Cl_o}{r^2} \left(\frac{1}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right), \\
 A_2 &= \rho \left[C^2 + Cl_o^2 \left(3\frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + \frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) - \frac{Cl_o}{2} \left((12+3A)\frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + (4+B)\frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) + \frac{\partial C}{\partial t} l_o \right] \\
 &\quad + \frac{6\mu Cl_o}{r^2} \left(\frac{1}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right), \\
 A_3 &= \rho \left[C^2 + Cl_o^2 \left(\frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + 3\frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) - \frac{Cl_o}{2} \left((4+A)\frac{\dot{\lambda}_1^{(f)}}{\dot{\lambda}_1^{(f)}} + (12+3B)\frac{\dot{\lambda}_2^{(f)}}{\dot{\lambda}_2^{(f)}} \right) + \frac{\partial C}{\partial t} l_o \right] \\
 &\quad + \frac{6\mu Cl_o}{r^2} \left(\frac{1}{\dot{\lambda}_1^{(f)2}} + \frac{1}{\dot{\lambda}_2^{(f)2}} \right)
 \end{aligned}$$

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